

Threat Detection & Incident Response using Splunk Enterprise

Security Information and Event Management (SIEM) Laboratory

Prepared By

Ibrahim Khaleel

Project Type

Security Operations Center (SOC)

Detection Engineering

Threat Monitoring

Security Analytics

Software

Splunk Enterprise 10.4.x

Ubuntu Server 24.04.4 LTS

Windows 11 Pro

Kali Linux

Microsoft Sysmon

Splunk Universal Forwarder

Version

1.0

July 2026

Threat Detection & Incident Response using Splunk Enterprise.....	1
Security Information and Event Management (SIEM) Laboratory	1
1 Introduction	2
1.1 Overview.....	2
1.2 Project Objectives	2
2 Lab Architecture	3
2.1 System Components	3
2.2 Operating Systems	3
2.3 Network Architecture.....	3
2.4 Data Flow.....	4
3 Splunk Configuration	4
3.1 Splunk Enterprise	4
3.2 Windows Endpoint Configuration.....	5
3.3 Log Sources	5
3.4 Universal Forwarder.....	5
3.5 Sysmon.....	5
4. Detection Engineering.....	6
4.1 Overview.....	6
4.2 MITRE ATT&CK Coverage	6
5. Detection Use Case 1 – PowerShell Execution Detection	7
5.1 Objective	7
5.2 Attack Simulation	7
5.3 MITRE ATT&CK.....	7
5.4 SPL Query	8
5.5 Query Explanation	8
5.6 Alert.....	8
5.7 Screenshot.....	9
5.8 Analyst Notes	9

6. Detection Use Case 2 – Encoded PowerShell Detection	10
6.1 Objective	10
6.2 Attack Simulation	10
6.3 MITRE ATT&CK.....	10
6.4 SPL Query	10
6.5 Query Explanation	11
6.6 Alert.....	11
6.7 Screenshot.....	11
6.8 Analyst Notes	12
7. Detection Use Case 3 – Suspicious Parent–Child Process Detection	12
7.1 Objective	12
7.2 Attack Simulation	12
7.3 MITRE ATT&CK.....	12
7.4 SPL Query	13
7.5 Query Explanation	13
7.6 Alert.....	13
7.7 Screenshot.....	13
7.8 Analyst Notes	14
8. Detection Use Case 4 – Windows Authentication Brute Force Detection	14
8.1 Objective	14
8.2 Attack Simulation	15
8.3 MITRE ATT&CK Mapping	15
8.4 SPL Query	15
8.5 Query Explanation	16
8.6 Alert Configuration.....	16
8.7 Expected Output.....	16
8.8 Screenshot.....	16
8.9 Analyst Investigation	17
9. Detection Use Case 5 – TCP Port Scan Detection	17

9.1 Objective	17
9.2 Attack Simulation	18
9.3 MITRE ATT&CK Mapping	18
9.4 SPL Query	18
9.5 Query Explanation	19
9.6 Alert Configuration.....	19
9.7 Expected Output.....	19
9.8 Screenshot.....	19
9.9 Analyst Investigation	20
9.10 Alert Engineering.....	20
Overview	21
Configured Alerts	21
Alert Workflow	21
Screenshot.....	22
Summary.....	22
10. Dashboard Analysis	22
10.1 Overview	23
10.2 SOC Executive Dashboard.....	23
10.3 Endpoint Detection Dashboard	24
10.4 Authentication & Network Dashboard.....	25
11. Testing & Results	26
11.1 Overview	26
11.2 Detection Validation	26
11.3 Results Summary.....	27
12. Problems Faced & Troubleshooting.....	27
13. Conclusion.....	28
14. Future Improvements.....	28
15. References.....	29
Appendix A – Detection Summary.....	29

1 Introduction

1.1 Overview

Security Operations Centers (SOCs) rely on Security Information and Event Management (SIEM) platforms to collect, analyze, and correlate security logs from multiple systems. These platforms help security analysts identify suspicious activities, investigate incidents, and respond to potential threats in a centralized manner.

Splunk Enterprise is one of the most widely adopted SIEM platforms due to its powerful Search Processing Language (SPL), real-time dashboards, alerting capabilities, and extensive support for log collection from diverse data sources.

This project demonstrates the deployment of a Splunk Enterprise-based SOC laboratory using Ubuntu Server as the SIEM platform, Windows 11 as the monitored endpoint, and Kali Linux as the attack machine. The environment was configured to collect Windows Security Logs, Sysmon telemetry, and Windows Firewall logs through the Splunk Universal Forwarder. Custom SPL queries were developed to detect common attack techniques, generate alerts, and visualize security events through dashboards aligned with the MITRE ATT&CK framework.

1.2 Project Objectives

The objectives of this project are:

- Deploy Splunk Enterprise as the centralized SIEM platform.
- Configure Windows event log collection.

- Install and configure Microsoft Sysmon.
- Configure Splunk Universal Forwarder.
- Collect Windows Firewall logs.
- Develop custom SPL detection rules.
- Create scheduled alerts.
- Design SOC dashboards.
- Validate detections using simulated attacks.

2 Lab Architecture

2.1 System Components

The laboratory consists of three primary systems:

Machine	Role
Ubuntu Server	Splunk Enterprise SIEM
Windows 11	Monitored Endpoint
Kali Linux	Attack Machine

2.2 Operating Systems

Machine	Operating System
Splunk Server	Ubuntu Server 24.04.4 LTS
Endpoint	Windows 11 Pro
Attacker	Kali Linux

2.3 Network Architecture

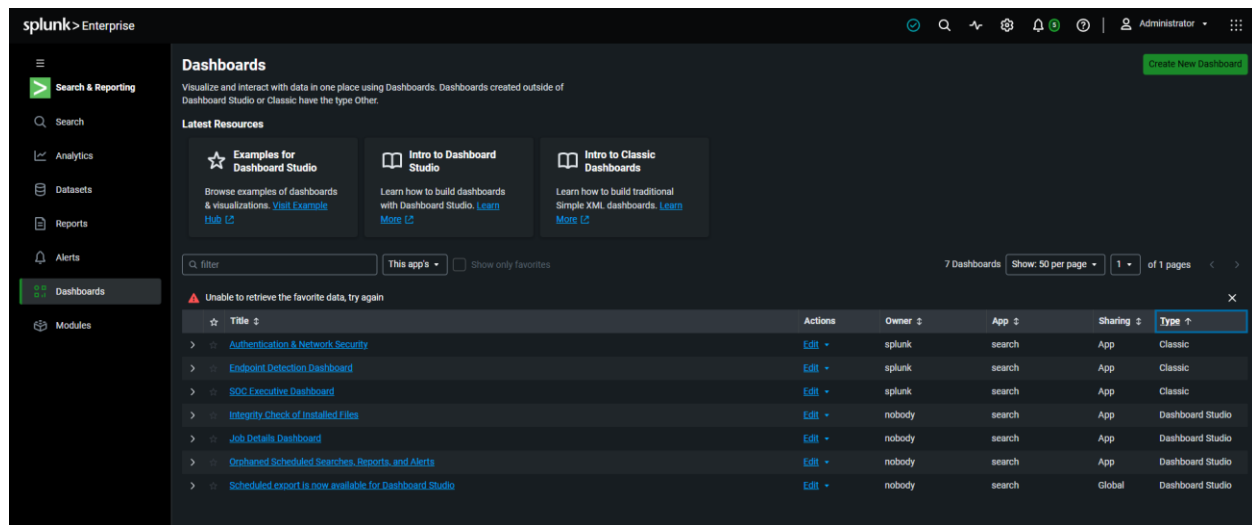


Figure 1. Laboratory Network Architecture

The Ubuntu Server hosts Splunk Enterprise and receives logs from the Windows endpoint using the Splunk Universal Forwarder. Kali Linux is used to simulate attack scenarios including PowerShell execution, brute-force authentication attempts, and TCP port scanning.

2.4 Data Flow

1. Windows 11 generates security events.
2. Sysmon records endpoint telemetry.
3. Windows Firewall logs network activity.
4. Universal Forwarder collects logs.
5. Logs are forwarded to Splunk Enterprise.
6. Splunk indexes the data.
7. SPL searches analyze events.
8. Alerts and dashboards present detection results.

3 Splunk Configuration

3.1 Splunk Enterprise

Splunk Enterprise was installed on Ubuntu Server and configured as the centralized SIEM platform responsible for ingesting, indexing, searching, and visualizing security logs.

The Splunk web interface was used to create dashboards, execute SPL queries, configure alerts, and validate detection results.

3.2 Windows Endpoint Configuration

The Windows endpoint was configured with:

- Microsoft Sysmon v15.21
- Splunk Universal Forwarder
- Windows Firewall Logging

This configuration ensured detailed endpoint telemetry and centralized log collection.

3.3 Log Sources

Source	Purpose
Security	Authentication events
Application	Application logs
System	Operating system events
Sysmon	Process monitoring
Firewall	Network activity

3.4 Universal Forwarder

The Splunk Universal Forwarder was configured to continuously monitor Windows Event Logs and Windows Firewall logs, securely forwarding them to the Splunk Enterprise server.

3.5 Sysmon

Sysmon enhanced endpoint visibility by recording:

- Process Creation
- Command Lines
- Parent Processes
- User Context
- Executable Path

These events formed the foundation of the custom detection rules implemented in this project.

4. Detection Engineering

4.1 Overview

Detection engineering is the process of designing and implementing detection logic to identify malicious activities from security logs. Instead of relying solely on built-in detection rules, custom Search Processing Language (SPL) queries were developed to detect common attack techniques using Windows Security Logs, Sysmon events, and Windows Firewall logs.

Each detection was tested through simulated attacks generated from the Kali Linux attack machine or the Windows endpoint. After validating the search results, scheduled alerts were configured, and the detections were incorporated into Splunk dashboards for continuous monitoring.

The detection engineering workflow used in this project is shown below.

Attack Simulation → Log Collection → SPL Query → Validation → Alert → Dashboard

4.2 MITRE ATT&CK Coverage

Detection	ATT&CK Technique	Tactic
PowerShell Execution	T1059.001	Execution
Encoded PowerShell	T1059.001	Execution
Parent–Child Process	T1204	Execution
Windows Brute Force	T1110	Credential Access
TCP Port Scan	T1046	Discovery

5. Detection Use Case 1 – PowerShell Execution Detection

5.1 Objective

The objective of this detection is to identify every instance of PowerShell execution on the monitored Windows endpoint. PowerShell is widely used by system administrators but is also a common tool abused by attackers for malware execution, persistence, and lateral movement.

Monitoring PowerShell activity provides analysts with visibility into potentially malicious script execution.

5.2 Attack Simulation

PowerShell was launched manually on the Windows endpoint using several commands to generate Sysmon Process Creation events. These events were forwarded to Splunk Enterprise using the Splunk Universal Forwarder.

5.3 MITRE ATT&CK

Field	Value
Technique	T1059.001
Name	PowerShell
Data Source	Sysmon
Event	EventCode 1

5.4 SPL Query

```
source="WinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| rex field=_raw "<Data Name='Image'>(?!<Image>[^\<]+)</Data>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^\<]*)</Data>"
| search Image="*powershell.exe"
| table _time Computer User Image CommandLine
| sort -_time
```

5.5 Query Explanation

The query retrieves Sysmon Process Creation events and extracts the executable path and command-line arguments from the XML event. It filters for PowerShell executions and displays relevant information such as execution time, hostname, user account, executable path, and command line.

5.6 Alert

Alert Name

PowerShell Execution Detection

Trigger Condition

Number of Results > 0

Severity

Medium

5.7 Screenshot

The screenshot displays the Splunk Enterprise web interface. On the left is a navigation sidebar with options like Search & Reporting, Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The main area is titled 'New Search' and contains a search query: `source="WinEventLog:Microsoft-Windows-Sysmon/Operational" | rex field=_raw "(?i)(.*)powershell.exe" | search Image="powershell.exe" | table _time Computer User Image CommandLine`. Below the query, it shows '1,283 events' and a time range of '7/26/26 3:00:00.000 PM to 7/27/26 3:20:49.000 PM'. The search results are displayed in a table with columns: _time, Computer, User, Image, and CommandLine. The table shows multiple entries of PowerShell execution from various computers and users.

_time	Computer	User	Image	CommandLine
2026-07-27 08:09:12.715			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:09:18.961			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:09:12.696			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:08:18.972			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:07:12.717			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:07:18.959			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:06:12.706			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:06:18.954			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:05:12.698			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:05:18.957			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:04:12.715			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:04:18.943			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:03:12.701			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:03:18.931			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:02:12.681			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:02:18.927			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:01:12.691			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:01:18.924			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:00:12.693			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	
2026-07-27 08:00:18.912			C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	

Figure 2. PowerShell Execution Detection in Splunk

5.8 Analyst Notes

- Verify the executing user.
- Review the complete command line.
- Determine whether execution is expected.
- Correlate with other endpoint events.

6. Detection Use Case 2 – Encoded PowerShell Detection

6.1 Objective

Attackers often use Base64-encoded PowerShell commands to hide malicious payloads from administrators and security tools. This detection identifies PowerShell processes launched using the -EncodedCommand or -enc parameter.

6.2 Attack Simulation

An encoded PowerShell command was executed on the Windows endpoint. Sysmon captured the process creation event, which was then forwarded to Splunk for analysis.

6.3 MITRE ATT&CK

Field	Value
Technique	T1059.001
Tactic	Execution
Data Source	Sysmon

6.4 SPL Query

```
source="WinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| rex field=_raw "<Data Name='Image'>(?!<Image>[^\<]+)</Data>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^\<]*)</Data>"
| where like(lower(Image), "%powershell%")
| where match(lower(CommandLine), "-enc(odedcommand)?")
```

```
| table _time Computer User Image CommandLine  
| sort -_time
```

6.5 Query Explanation

The search identifies Sysmon Process Creation events where PowerShell is executed with encoded command-line arguments. This technique is commonly used to obfuscate malicious scripts.

6.6 Alert

Alert Name

Obfuscated PowerShell Command Detection

Severity

High

6.7 Screenshot

The screenshot displays the Splunk Enterprise interface. On the left is a navigation sidebar with options like Search & Reporting, Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The main panel shows an alert titled "Encoded PowerShell Execution Detection". Above the alert, a search query is visible: `source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 | rex field=_raw "Data Name='Image'>{<Image>{<*>}</Data>" | rex field=_raw "Data Name='CommandLine'>{<CommandLine>{<*>}</Data>" | where like(lower(Image), "powershell") | where match(lower(CommandLine), "-enc(.*command).?") | table _time Computer User Image CommandLine | sort -_time`. Below the query, it indicates "5 events (7/26/20 3:00:00.000 PM to 7/27/20 3:09:15.000 PM)". The event list shows five entries, all with the same details: `C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe` as the Image and `"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA` as the CommandLine.

_time	Computer	User	Image	CommandLine
2020-07-27 12:31:49.337	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA
2020-07-27 11:57:26.436	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA
2020-07-27 07:06:58.601	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA
2020-07-27 07:05:33.856	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA
2020-07-27 07:01:07.356	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d6b0AGAY@IAGA

Figure 3. Encoded PowerShell Detection

6.8 Analyst Notes

- Review the encoded command.
- Decode the Base64 string if necessary.
- Investigate the parent process.
- Check for subsequent network activity.

7. Detection Use Case 3 – Suspicious Parent–Child Process Detection

7.1 Objective

Legitimate Windows processes generally follow predictable parent–child relationships. Attackers frequently exploit abnormal process chains such as `cmd.exe` spawning `powershell.exe` to execute malicious commands.

This detection identifies suspicious parent–child process relationships.

7.2 Attack Simulation

The Windows Command Prompt was used to launch PowerShell, generating Sysmon Process Creation events with parent and child process information.

7.3 MITRE ATT&CK

Field	Value
Technique	T1204
Name	User Execution
Data Source	Sysmon

7.4 SPL Query

```
source="WinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| rex field=_raw "<Data Name='Image'>(?!<Image>[^\<]+)</Data>"
| rex field=_raw "<Data Name='ParentImage'>(?!<ParentImage>[^\<]+)</Data>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^\<]*)</Data>"
| where (
  (like(lower(ParentImage), "%cmd.exe%") AND like(lower(Image), "%powershell.exe%"))
OR
  (like(lower(ParentImage), "%powershell.exe%") AND like(lower(Image), "%cmd.exe%"))
OR
  (like(lower(ParentImage), "%powershell.exe%") AND like(lower(Image),
"%powershell.exe%")))
)
| table _time ParentImage Image CommandLine
| sort -_time
```

7.5 Query Explanation

The query extracts the parent process, child process, and command-line arguments from Sysmon Process Creation events. It detects suspicious relationships commonly associated with malware execution and attacker activity.

7.6 Alert

Alert Name

Suspicious Parent–Child Process Detection

Severity

High

7.7 Screenshot

The screenshot shows the Splunk Enterprise interface with a search for 'Suspicious Parent-Child Process Detection'. The search results are displayed in a table with columns: _time, ParentImage, Image, and CommandLine. The results show several events where a parent process (C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe) spawned a child process (C:\Windows\System32\cmd.exe) and executed a command line (C:\Windows\System32\cmd.exe /c powershell.exe).

_time	ParentImage	Image	CommandLine
2020-07-27 12:43:46.357	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d4WAGAY0B1AGAA
2020-07-27 12:43:37.855	C:\Windows\System32\cmd.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe
2020-07-27 12:43:37.885	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c powershell.exe
2020-07-27 11:57:26.438	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d4WAGAY0B1AGAA
2020-07-27 11:57:23.755	C:\Windows\System32\cmd.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe
2020-07-27 11:57:23.789	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c powershell.exe
2020-07-27 11:55:47.384	C:\Windows\System32\cmd.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe
2020-07-27 11:55:47.239	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c powershell.exe
2020-07-27 07:46:59.681	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d4WAGAY0B1AGAA
2020-07-27 07:45:53.856	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand d4WAGAY0B1AGAA
2020-07-27	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand

Figure 4. Parent-Child Process Detection

7.8 Analyst Notes

- Verify the parent process.
- Review the child process.
- Examine the command line.
- Correlate with authentication events.
- Check for related network connections.

8. Detection Use Case 4 – Windows Authentication Brute Force Detection

8.1 Objective

Brute-force attacks are commonly used by attackers to gain unauthorized access by repeatedly attempting different username and password combinations. Monitoring failed authentication events helps identify potential password guessing attacks before a successful compromise occurs.

The objective of this detection is to identify repeated failed logon attempts originating from the same source IP address using Windows Security logs.

8.2 Attack Simulation

A brute-force attack was simulated from the Kali Linux machine using the **Hydra** tool against the Windows Remote Desktop Protocol (RDP) service. Multiple invalid login attempts generated Windows Security Event **4625 (Failed Logon)**, which was collected by Splunk Enterprise through the Universal Forwarder.

8.3 MITRE ATT&CK Mapping

Field	Value
Technique	T1110
Technique Name	Brute Force
Tactic	Credential Access
Log Source	Windows Security Log

8.4 SPL Query

```
index=* EventCode=4625
| eval Source_IP=coalesce(Source_Network_Address, SourceIp, IPAddress)
| stats count AS failed_attempts values(Account_Name) AS targeted_accounts by
Source_IP
| where failed_attempts >= 5
| sort - failed_attempts
```

8.5 Query Explanation

The query retrieves Windows failed logon events (EventCode=4625), normalizes the source IP address using the `coalesce()` function, groups events by source IP, and counts the number of failed authentication attempts. If the number of failed attempts reaches five or more, the event is flagged as a potential brute-force attack.

8.6 Alert Configuration

Property	Value
Alert Name	Windows Authentication Brute Force Detection
Search Type	Scheduled
Trigger Condition	Number of Results > 0
Severity	High

8.7 Expected Output

The detection provides:

- Source IP Address
- Number of Failed Attempts
- Targeted User Accounts
- Timestamp

8.8 Screenshot

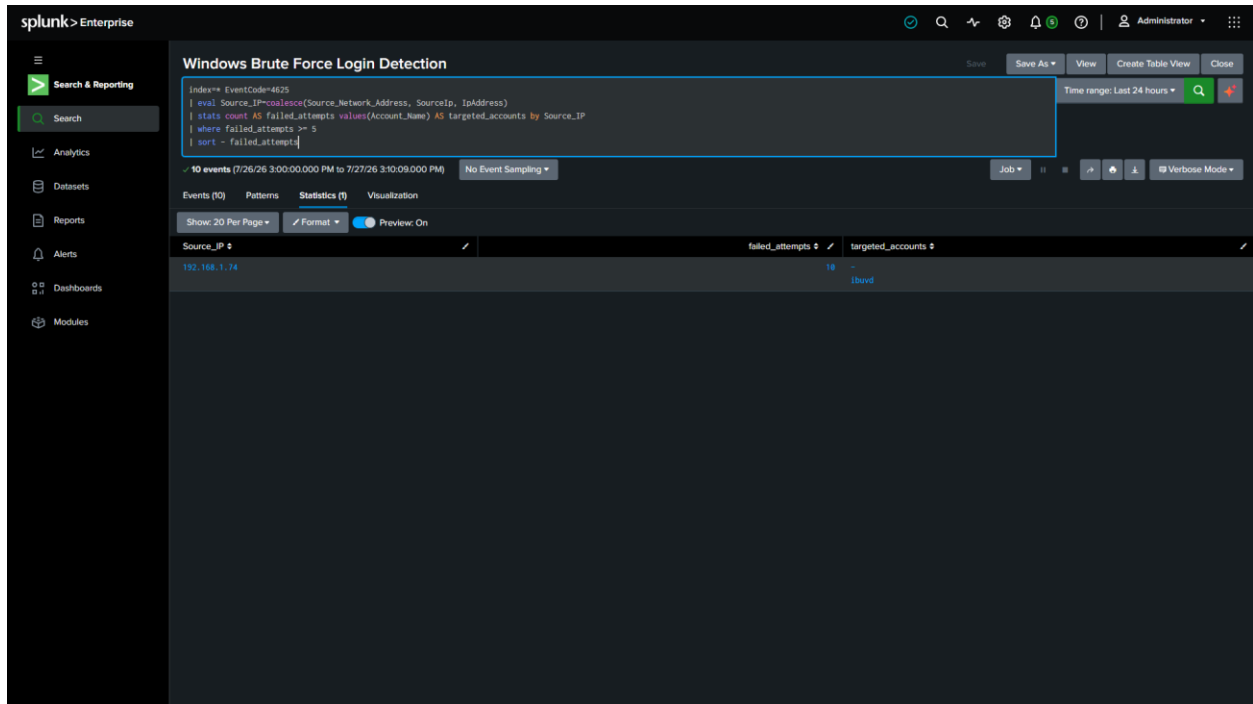


Figure 5. Windows Authentication Brute Force Detection

8.9 Analyst Investigation

After receiving the alert, the analyst should:

- Verify whether the source IP is internal or external.
- Review targeted user accounts.
- Check for successful logon events (EventCode 4624).
- Correlate the activity with Sysmon and firewall logs.
- Block the source IP if confirmed malicious.

9. Detection Use Case 5 – TCP Port Scan Detection

9.1 Objective

Reconnaissance is typically the first stage of an attack. Tools such as **Nmap** are used to identify active hosts, open ports, and exposed services before exploitation.

The objective of this detection is to identify TCP port scanning activity by analyzing Windows Firewall logs.

9.2 Attack Simulation

An Nmap TCP scan was executed from the Kali Linux machine against the Windows endpoint. Windows Firewall logged the incoming connection attempts in `pfirewall.log`, which was monitored by the Splunk Universal Forwarder and forwarded to Splunk Enterprise.

9.3 MITRE ATT&CK Mapping

Field	Value
Technique	T1046
Technique Name	Network Service Discovery
Tactic	Discovery
Data Source	Windows Firewall Log

9.4 SPL Query

```
index=main sourcetype=pfirewall
| rex field=_raw
"(<action>ALLOW|DROP)\s+(<protocol>\S+)\s+(<src_ip>\S+)\s+(<dest_ip>\S+)\s+(<src
```

```

_port>\d+|-)\s+(?<dest_port>\d+|-)"
| search protocol=TCP
| stats dc(dest_port) AS unique_ports values(dest_port) AS scanned_ports count AS
total_connections by src_ip
| where unique_ports >= 3
| rename src_ip AS "Source IP"
| rename unique_ports AS "Unique Ports Scanned"
| rename scanned_ports AS "Ports"
| rename total_connections AS "Connection Attempts"

```

9.5 Query Explanation

The query extracts network fields from the Windows Firewall log using regular expressions. It filters TCP traffic, counts the number of unique destination ports contacted by each source IP address, and identifies systems that exceed the configured threshold. Such behavior is characteristic of network reconnaissance tools such as Nmap.

9.6 Alert Configuration

Property	Value
Alert Name	Network Reconnaissance Detection
Search Type	Scheduled
Trigger Condition	Number of Results > 0
Severity	Medium

9.7 Expected Output

The search returns:

- Source IP
- Number of Unique Ports Scanned
- List of Scanned Ports
- Total Connection Attempts

This information enables analysts to quickly identify and investigate reconnaissance activity.

9.8 Screenshot

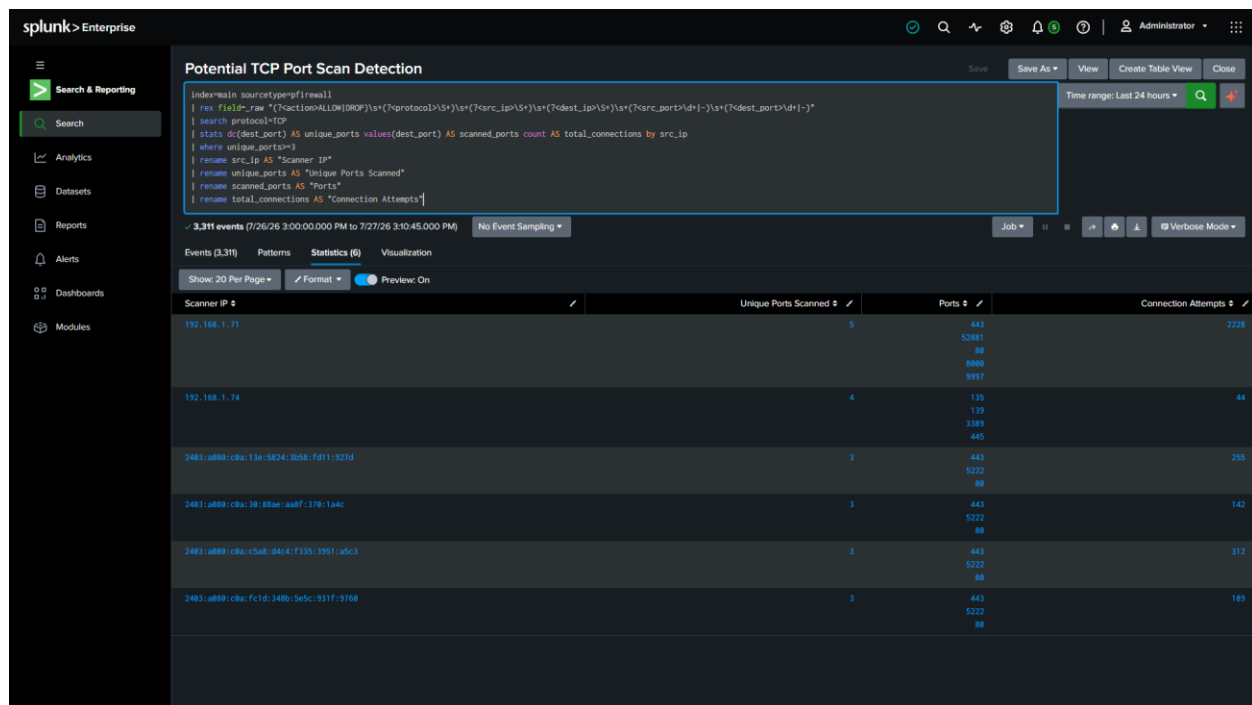


Figure 6. TCP Port Scan Detection

9.9 Analyst Investigation

When a port scan is detected, the analyst should:

- Determine whether the source IP belongs to an approved vulnerability scanner.
- Review the scanned ports.
- Check for subsequent exploitation attempts.
- Correlate with authentication and Sysmon events.
- Monitor or block the source host if unauthorized.

9.10 Alert Engineering

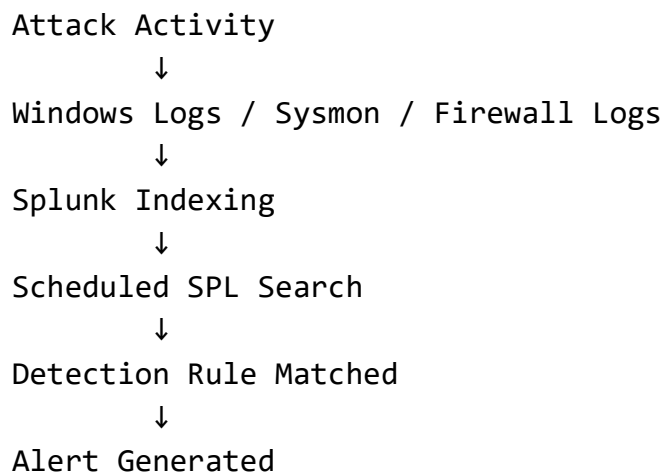
Overview

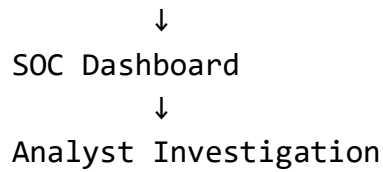
Once each SPL query was validated, scheduled alerts were created in Splunk Enterprise to continuously monitor incoming events. Alerting automates the detection process by notifying analysts whenever suspicious activity matches predefined detection logic.

Configured Alerts

Alert Name	Severity	Status
PowerShell Execution Detection	Medium	Enabled
Obfuscated PowerShell Command Detection	High	Enabled
Suspicious Parent–Child Process Detection	High	Enabled
Windows Authentication Brute Force Detection	High	Enabled
Network Reconnaissance Detection	Medium	Enabled

Alert Workflow





Screenshot

The screenshot displays the 'Searches, Reports, and Alerts' section of the Splunk Enterprise web interface. It shows a list of six configured alerts with columns for Name, Actions, Type, Next Scheduled Time, Display View, Owner, App, Alerts, Sharing, and Status. The alerts are: Potential TCP Port Scan Detection, Encoded PowerShell Execution Detection, Windows Brute Force Login Detection, Suspicious Parent-Child Process Detection, PowerShell Execution Alert, and PowerShell Execution Detection. All alerts are set to 'Alert' type and are 'Enabled'.

Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status
Potential TCP Port Scan Detection Detects a source IP attempting connections to multiple TCP destination ports within the monitoring window, indicating possible reconnaissance or port scanning activity.	Edit Run View Recent	Alert	2026-07-27 15:05:00 UTC	none	splunk	search	17	App	✓ Enabled
Encoded PowerShell Execution Detection Detects PowerShell execution using EncodedCommand or whoami command technique used to obfuscate malicious commands.	Edit Run View Recent	Alert	2026-07-27 15:05:00 UTC	none	splunk	search	96	App	✓ Enabled
Windows Brute Force Login Detection Detects repeated Windows login attempts from the same source IP, indicating a possible brute force attack.	Edit Run View Recent	Alert	2026-07-27 15:09:00 UTC	none	splunk	search	58	App	✓ Enabled
Suspicious Parent-Child Process Detection Detects suspicious parent and process relationships involving PowerShell and Command Prompt that may indicate script execution or malicious command chaining.	Edit Run View Recent	Alert	2026-07-27 15:05:00 UTC	none	splunk	search	36	App	✓ Enabled
PowerShell Execution Alert Detects PowerShell execution using Splunk Event ID 1.	Edit Run View Recent	Alert	2026-07-27 15:05:00 UTC	none	splunk	search	8	App	✓ Enabled
PowerShell Execution Detection Detects Windows PowerShell process creation using Splunk Event ID 1.	Edit Run	Report	none	none	splunk	search	0	Private	✓ Enabled

Figure 7. Configured Alerts in Splunk Enterprise

Summary

At the end of this phase, the SIEM environment was capable of detecting and alerting on five distinct attack scenarios:

Detection	Status
PowerShell Execution	Implemented
Obfuscated PowerShell	Implemented
Parent–Child Process	Implemented
Windows Authentication Brute Force	Implemented

10. Dashboard Analysis

10.1 Overview

To improve security monitoring and incident analysis, three dashboards were developed in Splunk Enterprise. These dashboards provide real-time visibility into endpoint activity, authentication events, and custom detection rules. They help SOC analysts quickly identify suspicious behavior, prioritize investigations, and monitor the overall security posture of the environment.

10.2 SOC Executive Dashboard

Purpose

The SOC Executive Dashboard provides a high-level summary of security events and detections across the monitored environment. It is intended for SOC managers and analysts who require an overview of current security activities.

Key Widgets

- Total Process Events
- Detection Rules Triggered
- Failed Logons
- Detected Port Scanners
- Encoded PowerShell Alerts
- Recent Security Detections
- Event Trend Graph

Screenshot

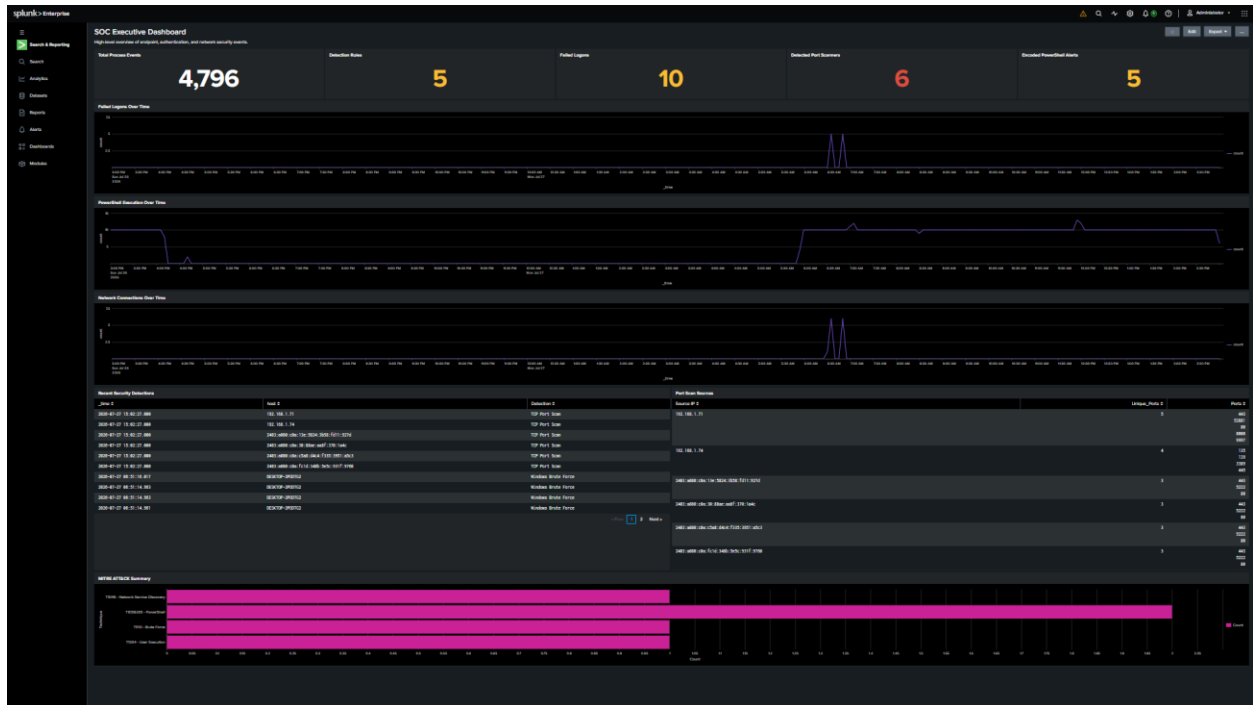


Figure 8. SOC Executive Dashboard

10.3 Endpoint Detection Dashboard

Purpose

This dashboard focuses on endpoint telemetry collected through Sysmon. It allows analysts to investigate process execution, PowerShell activity, and suspicious parent-child process relationships.

Key Widgets

- PowerShell Activity
- Encoded PowerShell Events
- Parent-Child Process Events
- Recent Process Executions

- Process Relationship Table

Screenshot

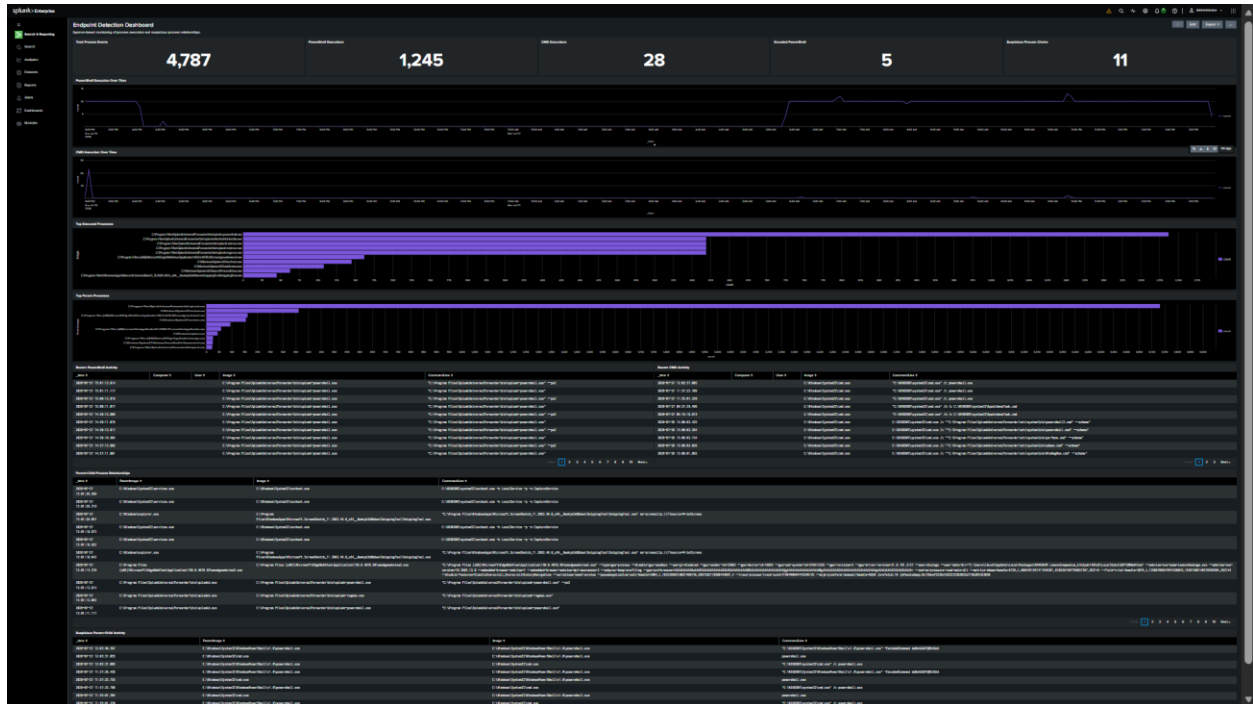


Figure 9. Endpoint Detection Dashboard

10.4 Authentication & Network Dashboard

Purpose

This dashboard monitors authentication activity and network reconnaissance attempts. It provides visibility into failed logons, successful logons, and TCP port scanning activity.

Key Widgets

- Failed Logons
- Successful Logons
- Port Scan Summary
- Network Connection Trends
- Recent Authentication Events

Screenshot

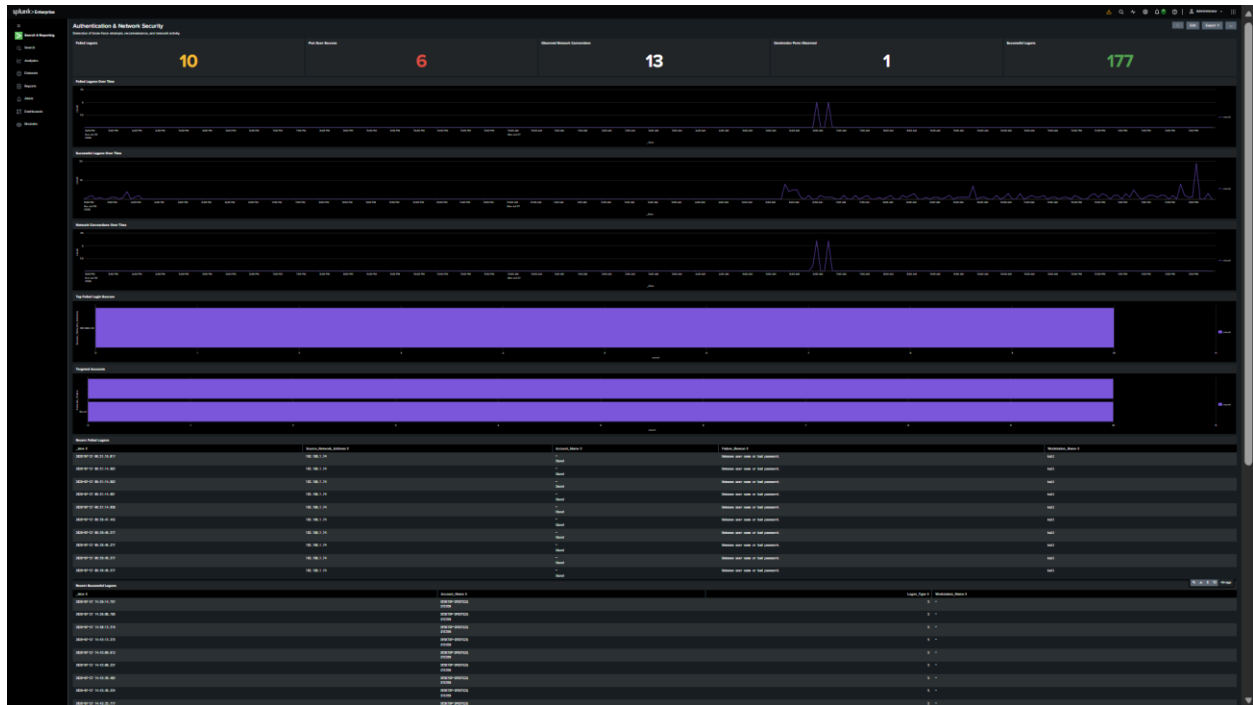


Figure 10. Authentication & Network Security Dashboard

11. Testing & Results

11.1 Overview

After configuring Splunk Enterprise and implementing the detection rules, controlled attack simulations were performed to verify that all detections, alerts, and dashboards functioned correctly.

Each attack generated the expected telemetry, which was successfully ingested, analyzed, and visualized within Splunk.

11.2 Detection Validation

Detection	Attack Simulated	Alert Generated	Status
PowerShell Execution	Manual PowerShell	Yes	Successful
Encoded PowerShell	Base64 Command	Yes	Successful
Parent-Child Process	cmd.exe → powershell.exe	Yes	Successful
Windows Brute Force	Hydra RDP Attack	Yes	Successful
TCP Port Scan	Nmap Scan	Yes	Successful

11.3 Results Summary

The project successfully demonstrated:

- Centralized log collection using Splunk Enterprise.
- Endpoint monitoring through Sysmon.
- Windows Firewall log analysis.
- Custom SPL query development.
- Scheduled alert generation.
- Dashboard visualization.
- MITRE ATT&CK-aligned detection engineering.

12. Problems Faced & Troubleshooting

During the implementation of the Splunk SOC laboratory, several technical issues were encountered and resolved.

Problem	Resolution
Sysmon fields were not extracted automatically	Used rex to extract XML fields such as Image, ParentImage, and CommandLine.
Universal Forwarder connection issues	Verified network connectivity, receiving port configuration, and firewall settings.
Windows Firewall logs not parsing correctly	Created custom field extraction using rex expressions.

Confusion between EventID and EventCode	Used EventCode, which is the normalized field for Sysmon events in Splunk.
Missing detection results	Verified log ingestion, corrected SPL queries, and tested with controlled attack simulations.

These troubleshooting steps improved the reliability of the SIEM environment and ensured that all detection use cases operated correctly.

13. Conclusion

This project successfully implemented a Security Information and Event Management (SIEM) laboratory using Splunk Enterprise. The environment consisted of an Ubuntu Server hosting Splunk Enterprise, a Windows 11 endpoint configured with Sysmon and the Splunk Universal Forwarder, and a Kali Linux machine used to simulate attack scenarios.

Five custom detection rules were developed using Splunk Search Processing Language (SPL) to identify PowerShell execution, obfuscated PowerShell commands, suspicious parent-child process relationships, Windows authentication brute-force attacks, and TCP port scanning activity. Each detection was validated through controlled testing, configured as a scheduled alert, and incorporated into operational dashboards.

The project demonstrates practical experience in SIEM deployment, log management, detection engineering, alert configuration, dashboard development, and threat monitoring. It also highlights the importance of endpoint telemetry and structured log analysis in identifying malicious activities within an enterprise environment.

Overall, this laboratory provides a realistic SOC environment that reflects common tasks performed by security analysts and serves as a valuable foundation for further learning in incident response and threat hunting.

14. Future Improvements

Future enhancements that can further strengthen this laboratory include:

- Integration with VirusTotal for file reputation analysis.
- Active Directory log monitoring.
- Integration of network sensors such as Suricata or Zeek.
- Threat intelligence feed integration.
- User and Entity Behavior Analytics (UEBA).
- Automated incident response using SOAR platforms.
- Detection of ransomware behaviors.
- Collection of logs from multiple Windows endpoints.

15. References

1. Splunk Enterprise Documentation
2. Splunk Search Processing Language (SPL) Reference
3. Microsoft Sysmon Documentation
4. Microsoft Windows Security Auditing Documentation
5. MITRE ATT&CK Framework
6. Nmap Reference Guide
7. THC Hydra Documentation

Appendix A – Detection Summary

Detection	Log Source	MITRE Technique	Alert
PowerShell Execution	Sysmon	T1059.001	Yes
Obfuscated PowerShell	Sysmon	T1059.001	Yes
Parent–Child Process	Sysmon	T1204	Yes
Windows Brute Force	Security Log	T1110	Yes
TCP Port Scan	Windows Firewall	T1046	Yes